Summarize

Appendix C. Trade-Off References

The primary focus of this book is trade-off analysis; to that end, we created a number of trade-off tables and figures in Part II to summarize trade-offs around a particular architecture concern. This appendix summarizes all the trade-off tables and figures for easy reference:

Figure 6-25, “Relational databases rated for various adoption characteristics”

Figure 6-26, “Key-value databases rated for various adoption characteristics”

Figure 6-27, “Document databases rated for various adoption characteristics”

Figure 6-28, “Column family databases rated for various adoption characteristics”

Figure 6-30, “Graph databases rated for various adoption characteristics”

Figure 6-31, “New SQL databases rated for various adoption characteristics”

Figure 6-32, “Cloud native databases rated for various adoption characteristics”

Figure 6-33, “Time-series databases rated for various adoption characteristics”

Table 8-1, “Trade-offs for the code replication technique”

Table 8-2, “Trade-offs for the shared library technique”

Table 8-3, “Trade-offs for the shared service technique”

Table 8-4, “Trade-offs for the Sidecar pattern / service mesh technique”

Table 9-1, “Joint ownership table split technique trade-offs”

Table 9-2, “Joint ownership data-domain technique trade-offs”

Table 9-3, “Joint ownership delegate technique trade-offs”

Table 9-4, “Joint ownership service consolidation technique trade-offs”

Table 9-5, “Background synchronization pattern trade-offs”

Table 9-6, “Orchestrated request-based pattern trade-offs”

Table 9-7, “Event-based pattern trade-offs”

Table 10-1, “Trade-offs for the Interservice Communication data access pattern”

Table 10-2, “Trade-offs for the Column Schema Replication data access pattern”

Table 10-3, “Trade-offs associated with the replicated caching data access pattern”

Table 10-4, “Trade-offs associated with the data domain data access pattern”

Table 11-1, “Trade-offs for orchestration”

Table 11-2, “Trade-offs for the Front Controller pattern”

Table 11-3, “Stateless choreography trade-offs”

Table 11-4, “Stamp coupling trade-offs”

Table 11-5, “Trade-offs for the choreography communication style”

Table 11-6, “Trade-off between orchestration and choreography for ticket workflow”

Table 11-7, “Updated trade-offs between orchestration and choreography for ticket workflow”

Table 11-8, “Final trade-offs between orchestration and choreography for ticket workflow”

Table 12-11, “Trade-offs associated with state management rather than atomic distributed transactions with compensating updates”

Table 12-12, “Trade-offs associated with atomic distributed transactions and compensating updates”

Table 13-1, “Trade-offs for strict contracts”

Table 13-2, “Trade-offs for loose contracts”

Table 13-3, “Trade-offs for consumer-driven contracts”

Table 13-4, “Trade-offs for stamp coupling”

Table 14-1, “Trade-offs for the Data Warehouse pattern”

Table 14-2, “Trade-offs for the Data Lake pattern”

Table 14-3, “Trade-offs for the Data Mesh pattern”

Table 15-2, “Consolidated comparison of dynamic coupling patterns”

Table 15-3, “Trade-offs between synchronous and asynchronous communication for credit card processing”

Table 15-4, “Trade-offs between point-to-point versus publish-and-subscribe messaging”